ORBITAL

Operational Reusable Backend for Integrated Trajectory and Logistics

Technical Report

Abstract

ORBITAL is a modular, domain-agnostic mission planning framework for constraint-critical aerospace systems. Traditional planners are tightly coupled to specific vehicle domains, limiting reuse and increasing engineering complexity. ORBITAL separates optimization mechanics from domain-specific physics, enabling a reusable backend capable of generating executable mission plans across heterogeneous domains, including fixed-wing UAV trajectory planning and LEO CubeSat scheduling.

This document summarizes the mathematical formulation, architecture, optimization method, physics integration, robustness framework, and validation results.

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1. Problem Formulation

Mission planning is modeled as a constrained optimization problem.

Let:

- $x \in \mathbb{R}^n$ be decision variables
- $J(x)$ be objective cost
- $C_i(x)$ be constraint margins

We solve:

Minimize

$$J(x)$$

Subject to

$$C_i(x) \geq 0 \quad \text{for all hard constraints}$$

Margins quantify distance to violation, enforcing a core principle:

Feasibility precedes optimality.

Constraint satisfaction is evaluated independently of objective ranking.

Decision Variables

Decision variables encode controllable mission parameters, including:

- Velocity profiles
- Waypoint permutations
- Timing schedules
- Resource allocation
- Observation sequencing

Supported types: continuous, integer, discrete, and permutation.
The search space is high-dimensional and non-convex; no gradient assumptions are made.

Constraint Model

Constraints use margin-based evaluation:

- Margin ≥ 0 → feasible
- Margin < 0 → violation

Hard constraints (fuel ≥ 0, battery ≥ 0, bank limits, slew limits, geofencing, visibility thresholds) must never be violated.
Soft constraints incur penalties but do not invalidate feasibility.

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2. Objective and Penalty Formulation

All objectives are represented in cost space. Maximization terms are sign-inverted.

Total score:

$S = J(x) + w \cdot \Sigma (\max(0, -C_i(x)))^2$

Hard constraint feasibility is enforced independently of score comparison. Squared penalties amplify large violations while preserving smooth scaling.

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3. Architecture

ORBITAL consists of four decoupled layers:

1. Decision Variables
2. Simulation (domain physics)
3. Constraints (margin evaluation)
4. Objective (performance scoring)

The optimization kernel is domain-agnostic. Simulation modules plug into a shared evaluation loop:

1. Mutate decision variables
2. Construct executable plan
3. Run simulation

4. Evaluate margins
5. Score
6. Iterate

This separation enables cross-domain reuse without modifying planner logic.

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4. Domain Physics

Aircraft Domain

The UAV implementation includes:

- Wind-aware kinematics
- Bank-angle-constrained turn radius
- Energy tracking
- Geofence enforcement

Results:

- 27.6% fuel reduction vs baseline
- 0 hard constraint violations

Spacecraft Domain

The spacecraft module incorporates:

- Two-body dynamics

- J2 perturbation
- RK4 integration
- Inertial propagation in ECI
- GMST-based ECI → ECEF transformation
- Elevation-angle ground-station visibility

Orbital Dynamics

State propagation:

$\ddot{r} = -\mu / r^3 \cdot r + a\_J2$

J2 perturbation term:

$$a\_J2 = (3 J_2 \mu R\_E^2 / 2 r^5) \cdot \begin{bmatrix} x (5 z^2/r^2 - 1), \\ y (5 z^2/r^2 - 1), \\ z (5 z^2/r^2 - 3) \end{bmatrix}$$

This models Earth oblateness and improves LEO accuracy.

ECI → ECEF Transformation

Operational evaluation uses:

$r\_ECEF = R_3(\theta\_GMST) \cdot r\_ECI$

Rotation matrix:

$$R_3(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Velocity correction:

$v\_ECEF = R_3(\theta) \cdot v\_ECI - \omega \times r\_ECEF$

Separating inertial propagation from Earth-fixed evaluation ensures consistent access timing and ground-track computation.

Seven-day LEO results:

- 87.5% target coverage
- 93.8% data return
- 0 hard violations

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5. Optimization Strategy

ORBITAL employs mutation-based heuristic search with annealed mutation intensity.

Mutation Model

- Gaussian perturbations for continuous variables
- Uniform discrete sampling for categorical variables
- Swap mutations for permutations

Annealing Schedule

$\sigma_k = \sigma_0 \cdot \exp(-\alpha k)$

Where σ_0 is the initial mutation scale and α controls decay.

Strict improvement is required for acceptance. Feasible solutions dominate infeasible ones.

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6. Robustness

Uncertainty modeling includes:

- Wind ±30%
- Mass ±5%
- Drag ±10%

Monte Carlo evaluation (100+ seeds):

- 97% hard feasibility rate
- Deterministic reproducibility via seeded RNG

Risk-aware aggregation supports CVaR-style evaluation.

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7. Validation and Performance

UAV Mission

- 27.6% fuel reduction
- 100% hard feasibility

CubeSat 7-Day Mission

- 87.5% coverage
- 93.8% data return
- 97% Monte Carlo feasibility

Computational Performance

Scenario	Evaluations	Runtime (s)	Hard Feasibility

UAV	200	3.2	100%
CubeSat	200	5.8	97%

Runtime scales approximately linearly with evaluation count (single-core execution).

The optimization kernel required no modification between domains.

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8. Limitations and Outlook

Limitations:

- Heuristic search without deterministic optimality guarantees
- No large-scale constellation benchmarking
- Simplified environmental modeling
- No parallel evaluation

Future directions include parallel scenario evaluation, constellation coordination, adaptive penalty weighting, and higher-fidelity environmental models.

Conclusion

ORBITAL demonstrates that a reusable, domain-agnostic optimization kernel can generate executable mission plans across heterogeneous aerospace domains while enforcing strict feasibility discipline.

By separating decision variables, simulation physics, constraint evaluation, and objective scoring, ORBITAL provides a scalable foundation for structured aerospace mission optimization.